# AI for Medicine - Chest X-ray Classification

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## 1. Project Title

Automated Multi-Class Chest X-ray Classification for Respiratory Disease Detection Using Deep Learning

## 2. Problem Statement

In today's world, the growing global population and the increasing demand on healthcare systems have made fast and reliable medical diagnosis more important than ever. During extreme public health situations such as the COVID-19 pandemic, delays in diagnosis can place patients' lives at risk and increase pressure on hospitals and medical staff.

## 3. Objective of the Study

This project addresses the need for an automated deep learning-based system that can classify chest X-ray images. The idea is to develop and evaluate a deep learning model to perform this classification task. The goal is not to replace doctors, but to support faster screening, assist clinical decision-making, and help prioritize patients in situations where rapid action is required.

## 4. Dataset Description

This study uses the COVID-19 Radiography Database, a publicly available chest X-ray dataset from Kaggle. The dataset was downloaded directly in the notebook using KaggleHub to improve reproducibility. The Dataset folder is then explored to identify potential issues that could affect the reliability, and can lead to unreliable results and non-reproducible machine learning experiments.

No data harmonization method was applied, because the available metadata did not provide acquisition site or scanner information. The dataset originally contained 21,165 images. After cleaning, the dataset contained 21,111 images, and the class distribution remains imbalanced.

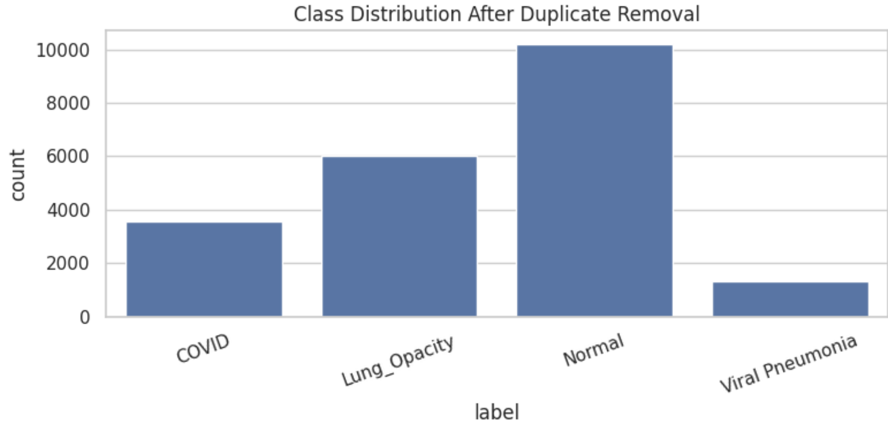


*Figure 1. Class distribution after exact duplicate removal.*

## 5. Data Processing

Image file paths and class labels were collected into a dataframe with three main fields: filepath, label, and label_encoded. All images were resized to 224 x 224 pixels, converted to three-channel RGB format, and normalized using ImageNet normalization. Basic data augmentation, including random horizontal flipping and small rotation, was applied only to the training data.

## 6. Avoiding Data Leakage

To reduce the risk of data leakage, exact duplicates were removed before splitting the dataset. The cleaned dataset was then divided using a stratified 80/20 train-test split, producing 16,888 training/development images and 4,223 independent test images.

The test set was kept untouched during model selection, hyperparameter tuning, and cross-validation. Stratified 5-fold cross-validation was performed only on the training/development set. A hash overlap check between the training and test sets returned zero overlap.

## 7. Machine Learning Pipeline

The project compared two deep learning models. The first model was a custom convolutional neural network trained from scratch and used as a baseline. The second model was MobileNetV2 pretrained on ImageNet and adapted for four-class classification through transfer learning.

The custom CNN contained 93,764 parameters, while MobileNetV2 contained 2,228,996 parameters after replacing the classifier head for four output classes. CrossEntropyLoss and AdamW optimization were used for training. Early stopping was applied to reduce unnecessary training when validation performance stopped improving.

Model selection was performed with stratified 5-fold cross-validation on the training/development set. Hyperparameter tuning was intentionally small and practical for Google Colab: the CNN baseline used a fixed configuration, while MobileNetV2 was tested with two learning rates and a fixed weight decay. The independent test set was not used during tuning.

## 8. Results

Stratified 5-fold cross-validation showed that MobileNetV2 clearly outperformed the custom CNN baseline. The CNN achieved a mean validation accuracy of 65.83%, while MobileNetV2 achieved the best mean validation accuracy of 94.72% using a learning rate of 0.0003 and weight decay of 0.0001.

| Model | Learning rate | Weight decay | Mean CV accuracy | Std. |
|---|---|---|---|---|
| CNN | 0.0010 | 0.0001 | 0.6583 | 0.0096 |
| MobileNetV2 | 0.0010 | 0.0001 | 0.9334 | 0.0106 |
| MobileNetV2 | 0.0003 | 0.0001 | 0.9472 | 0.0057 |

Based on cross-validation performance, MobileNetV2 with learning rate 0.0003 and weight decay 0.0001 was selected as the final model. After retraining on the full training/development set, it was evaluated once on the untouched independent test set.

| Metric | Value |
|---|---|
| Test accuracy | 0.9436 |
| Test loss | 0.1649 |
| Macro F1-score | 0.9539 |
| Weighted F1-score | 0.9431 |
| Macro AUC | 0.9938 |

Class-level performance was strongest for COVID-19 and Viral Pneumonia. Lung Opacity had the lowest recall, indicating that this class was the most difficult for the model to identify.

| Class | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| COVID | 0.9901 | 0.9790 | 0.9845 | 714 |
| Lung Opacity | 0.9672 | 0.8570 | 0.9088 | 1203 |
| Normal | 0.9149 | 0.9755 | 0.9442 | 2038 |
| Viral Pneumonia | 0.9604 | 0.9963 | 0.9780 | 268 |

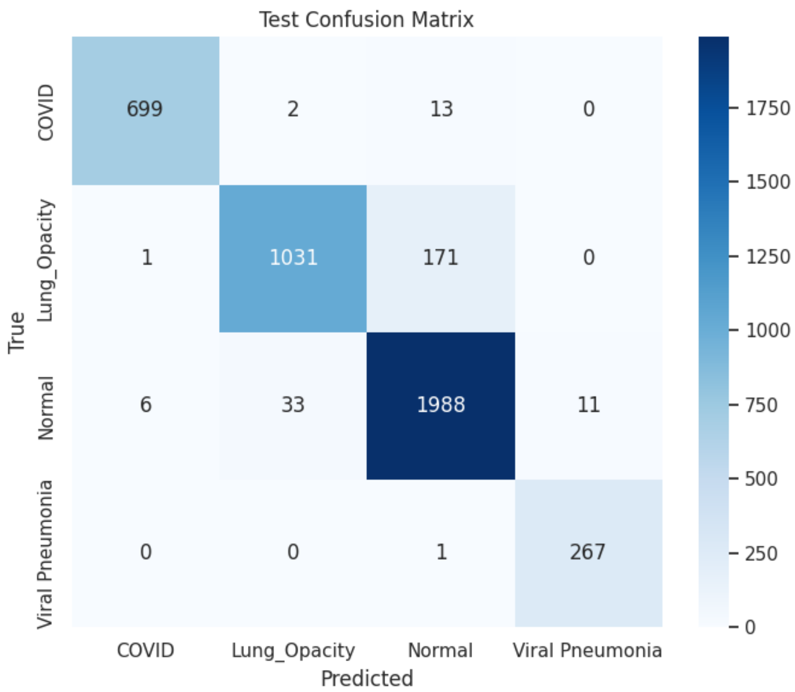


*Figure 2. Test confusion matrix for the selected MobileNetV2 model. Most errors occurred when Lung Opacity images were predicted as Normal.*

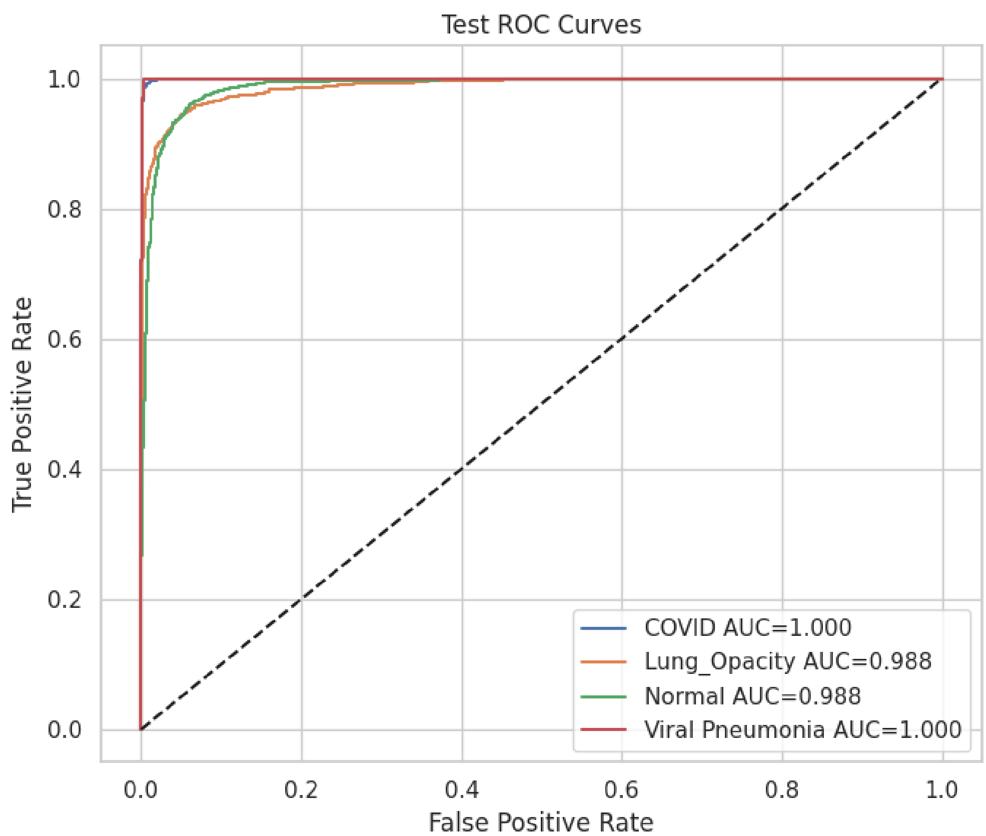


*Figure 3. One-vs-rest ROC curves on the independent test set. All classes achieved high AUC values, with macro AUC equal to 0.9938.*

## Grad-CAM Explainability

Grad-CAM was generated for qualitative explainability. The notebook produced one correctly classified example for each class and two misclassified examples. These visualizations are useful for inspecting model attention, but they should not be interpreted as clinical explanations.

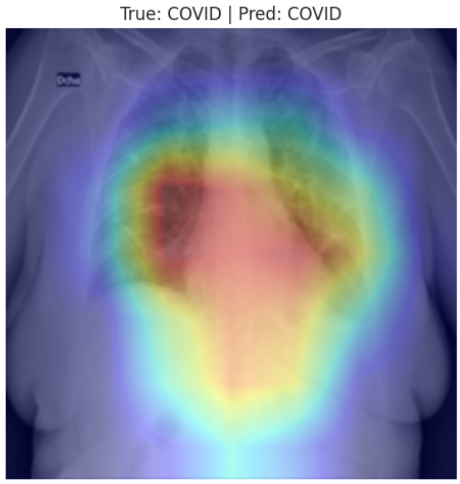


*Figure 4a. Correct COVID.*


*Figure 4b. Correct Lung Opacity.*


*Figure 4c. Correct Normal.*


*Figure 4d. Correct Viral Pneumonia.*

*Figure 4. Grad-CAM examples for correctly classified test images.*

## Grad-CAM Misclassified Examples


*Figure 5a. Normal predicted as Viral Pneumonia.*

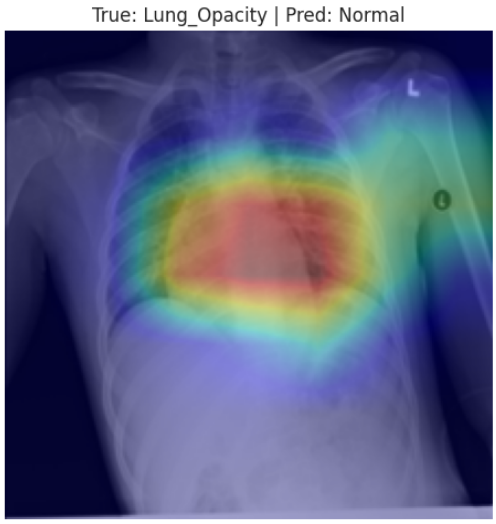


*Figure 5b. Lung Opacity predicted as Normal.*

The misclassified Lung Opacity example supports the quantitative result that Lung Opacity was the most challenging class for the model.

## 9. Discussion

The results show a clear advantage of transfer learning over training a small CNN from scratch. The custom CNN provided a useful baseline, but its lower cross-validation accuracy suggests that it was not able to learn sufficiently strong image features from the dataset alone. In contrast, MobileNetV2 achieved much higher and more stable performance, indicating that pretrained convolutional features were effective for this chest X-ray classification task.

Although the final MobileNetV2 model achieved strong overall performance, the class-level results show that Lung Opacity was the most challenging category. This may be due to the more subtle radiographic appearance of lung opacity and its possible visual overlap with normal chest X-rays. Therefore, the results highlight the importance of evaluating class-level metrics rather than relying only on overall accuracy.

The high AUC values indicate strong class separation in the model's predicted probabilities. However, AUC alone does not guarantee clinical reliability. External validation, calibration, and expert review would be required before any medical or deployment-oriented interpretation.

## 10. Conclusions and Future Work

This project demonstrated that a reproducible PyTorch-based pipeline can be used for automated four-class chest X-ray classification. MobileNetV2 clearly outperformed the custom CNN baseline and achieved strong performance on the independent test set.

Future work should focus on improving the classification of Lung Opacity, which showed the lowest recall. Additional steps could include stronger class imbalance handling, targeted data augmentation, lung-region-based preprocessing, probability calibration, and external validation on independent chest X-ray datasets.

Future work should also address data harmonization more explicitly. In this project, harmonization was not applied because acquisition site or scanner metadata were not available. A stronger future study should collect or use datasets with site, scanner, and acquisition-protocol information, then assess whether harmonization methods reduce domain shift and improve generalization across institutions.

Since MobileNetV2 is suitable for lightweight deployment, future work should also investigate Edge AI preparation through model compression, quantization, ONNX export, and inference-time evaluation on edge devices.

## 11. Ethics and Data Privacy

This project used a publicly available chest X-ray dataset from Kaggle. No personally identifiable patient information was accessed or processed directly. The work was conducted for educational and research purposes only.

The COVID-19 Radiography Database is subject to its original Kaggle dataset terms and any associated usage restrictions. The model should not be interpreted as a clinically validated diagnostic tool. Any real medical use would require expert clinical review, bias assessment, prospective validation, and appropriate regulatory evaluation.

## 12. Code and Reproducibility

The main notebook file is chest_xray.ipynb, developed for Google Colab with GPU acceleration. It contains the complete workflow from Kaggle dataset download to final test evaluation and Grad-CAM visualization. The notebook follows this workflow:

1. Install and import libraries.
2. Set random seeds and detect GPU.
3. Download the dataset from Kaggle.
4. Create a dataframe with filepath, label, and label_encoded.
5. Remove exact duplicate images before splitting.
6. Create a stratified 80/20 train-test split.
7. Keep the test set untouched during model selection.
8. Run stratified 5-fold cross-validation on the training set.
9. Compare CNN and MobileNetV2.
10. Select the best model from cross-validation.
11. Retrain the selected model on the full training set.
12. Evaluate once on the independent test set.
13. Generate confusion matrix, ROC curves, AUC scores, and Grad-CAM figures.

## 13. References

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